AWS Dashboard and EC2

Region, Availability Zone, EC2 and launching your first Ubuntu server: simple notes with steps and commands

Part	Topic	One-line idea
1	AWS Management Console	The web dashboard to control AWS
2	Region vs Availability Zone	Where your servers physically live
3	EC2	Rent a virtual server in the cloud
4	Launch your first Ubuntu instance	Step by step
5	Connect with SSH	Commands to copy and paste
6	First commands on the server	Show a working server to students
7	Manage and save money	Stop, terminate and billing safety
8	Quick quiz	Revise with answers

1. AWS Management Console (Dashboard)

The **AWS Management Console** is a **web application** where you manage everything in AWS with your mouse. Open it in a browser and sign in with your account.

What you can do	Examples
Use and manage services	EC2 (servers), S3 (storage), RDS (databases), Lambda (functions)
Watch usage and billing	See what you are spending and spot unusual costs
Set up security	Users, permissions, keys and firewalls

Parts of the dashboard

Part	What it does
Navigation bar (top)	Shows your account name and the **active region**. Gives access to global services like billing and support.
Service search bar	Type a service name (for example EC2) to open it quickly.
Resource Groups	Group related resources so you can manage them together.
Recently visited services	Quick list of services you used lately.
Build a solution	Quick links to common tasks, such as launching an instance or creating a database.

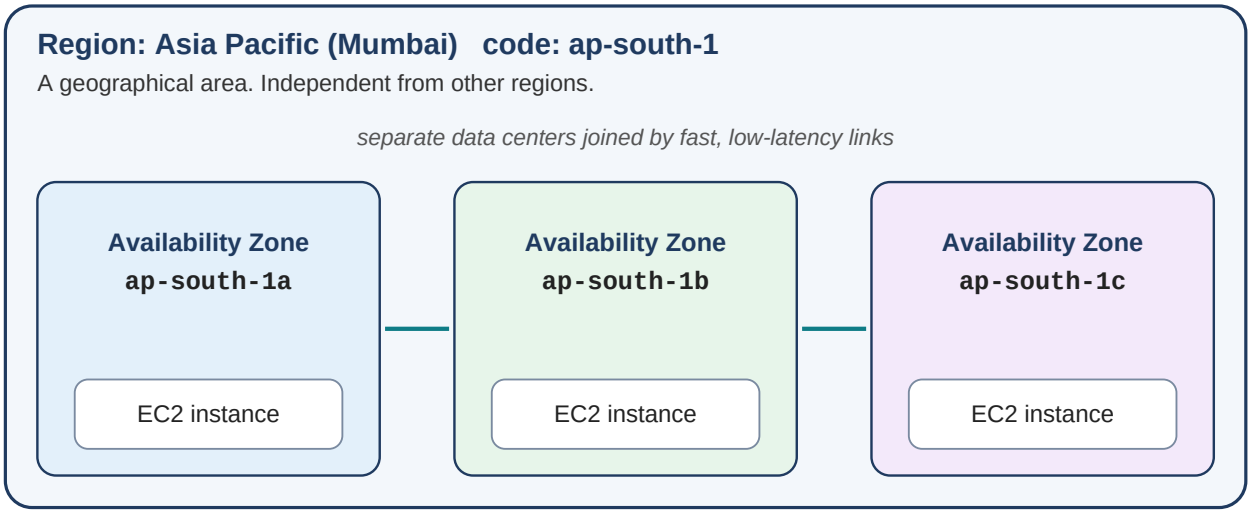
Note: AWS updates the look of the console often. The names may move, but the ideas stay the same. The **region selector** is in the top-right corner, next to your account name.

Best practices

- **Always check the active region** before you create anything, so you do not deploy in the wrong place.
- Use the **Billing and Cost Management** dashboard to track costs and spot unusual usage.
- Turn on **MFA** (multi-factor login) for your account, and do not use the root user for daily work.
- Create a **budget alert** (Billing, Budgets) so you get an email before costs grow.

2. Region vs Availability Zone (AZ)

AWS places your resources in **Regions** and **Availability Zones** to give **high availability**, **fault tolerance** and **low latency**.



	Region	Availability Zone (AZ)
What	A geographical location, e.g. US East (N. Virginia), Asia Pacific (Mumbai).	A separate data center (or group of them) inside a region.
Isolation	Each region is independent.	Each AZ is isolated, but joined to the other AZs of the region by fast links.
Count	Contains several AZs.	Mumbai (ap-south-1) has 3: ap-south-1a, 1b and 1c.
Code example	ap-south-1	ap-south-1a

Easy picture: A **Region** is like a **city**. An **Availability Zone** is like a **separate building** in that city, with its own power and network. If one building has a problem, the others keep working.

Why they matter

Benefit	Meaning
High availability	Run servers in two or more AZs. If one AZ fails, your app stays online.
Low latency	Choose a region close to your users so the network delay is small.
Cost	Prices can differ between regions for the same service.
Compliance	Some laws require data to stay in a certain country or region.

Remember: Choose a region by **distance to users, rules (compliance), and cost**. For users in India, **Asia Pacific (Mumbai) ap-south-1** is a common choice.

3. Introduction to EC2

Amazon EC2 (Elastic Compute Cloud) lets you **rent virtual servers** in the cloud. A server you launch is called an **instance**. It is the IaaS service from the virtualization notes: you choose the hardware size, and you control the operating system.

Feature	Meaning
Scalable	Add or remove instances (or change the size) as demand changes.
Customizable	Choose CPU, memory and storage by selecting an instance type.
Secure access	Use SSH key pairs and security groups (a firewall) to control who can connect.
Flexible pricing	Pay in different ways, shown below.

Instance type name: t2.micro

Part	Meaning
t	Family: general purpose, burstable. (m = general, c = compute, r = memory, g = GPU)
2	Generation (newer number = newer hardware).
micro	Size. micro = 1 vCPU and 1 GiB RAM. (small, medium, large get bigger.)

Pricing models

Model	How it works	Best for
On-Demand	Pay only while it runs (Linux is billed per second). No commitment.	Learning, short or unpredictable work
Reserved Instances / Savings Plans	Commit to 1 or 3 years and get a lower price.	Servers that run all the time
Spot Instances	Use spare AWS capacity at a big discount. You pay the current Spot price (no bidding now). AWS can take it back with a 2-minute warning.	Batch jobs, testing, work that can restart

Use cases

- Hosting web applications.
- Running batch jobs or machine learning models.
- Building development and testing environments.

Free tier: Free tier: the rules changed in July 2025. Accounts created **before 15 July 2025** get 750 hours a month of t2.micro (or t3.micro) for 12 months. Newer accounts get **free credits** instead (about $100 to $200), and EC2 usage uses up those credits. Always read the current **AWS Free Tier** page, and check which instance types show **Free tier eligible** in your console.

4. Launch Your First EC2 Instance (Ubuntu)

Note: Your notes describe the older step-by-step wizard (AMI, type, details, storage, tags, security group, review). The new console shows **all choices on one page**. The same choices are there, listed below in that page's order. Ubuntu 20.04 is now old, so choose **Ubuntu Server 22.04 LTS or 24.04 LTS**.

1. **Open EC2**
   Sign in, search for EC2, check the region (top right), click Launch instance
2. **Name and tags**
   Name: MyFirstInstance (this becomes the tag Key = Name)
3. **Choose the AMI (operating system image)**
   Ubuntu Server 22.04 LTS (marked Free tier eligible)
4. **Choose the instance type**
   t2.micro or t3.micro (1 vCPU, 1 GiB RAM, Free tier eligible)
5. **Create a key pair (login)**
   New key pair, name it, type RSA, format .pem, download once
6. **Network settings**
   Auto-assign public IP: Enable. New security group: allow SSH (22) from My IP
7. **Storage**
   Keep the default 8 GiB (gp3). Increase only if needed
8. **Review and Launch**
   Check the Summary panel, click Launch instance, then View all instances

Term	Simple meaning
AMI	Amazon Machine Image: a ready-made template with an operating system (like an installation disc).
Instance type	The size of the server: CPU and memory.
Key pair	A lock and key for login. AWS keeps the public key, you keep the private **.pem** file. You can download it only once.
Security group	A virtual firewall: it lists which traffic (port and source) may reach the instance.
Port 22	The port used by SSH.
Public IP	The address used to reach the instance from the Internet. It can change when you stop and start.
EBS volume	The virtual disk of the instance (8 GiB by default).

Wait for the instance to be ready: in **Instances**, the **Instance state** must be **Running** and the status checks should show **2/2 checks passed**. Then click the instance and copy its **Public IPv4 address**.

Best practice: In the security group, set the SSH source to **My IP**, not 0.0.0.0/0 (everyone). Opening SSH to the whole Internet invites attacks. Use an **Elastic IP** if you need a public IP that never changes.

5. Connect to the Instance with SSH

SSH (Secure Shell) logs you in to the server's terminal safely over the network. You need the **.pem** key file and the **public IP**.

Syntax	Meaning
`chmod 400 KEYFILE.pem`	Make the key readable only by you. SSH refuses keys that others can read.
`ssh -i KEYFILE.pem USER@PUBLIC_IP`	Connect. -i = identity (key) file. USER is the login name.
`exit`	Log out of the server.

COPY AND PASTE: Connect from Linux or macOS (replace the key name and IP)

```
cd ~/Downloads
chmod 400 MyKey.pem
ssh -i MyKey.pem ubuntu@PUBLIC_IP
```

Command	What it does (in simple words)
`cd ~/Downloads`	Go to the folder where the key was downloaded.
`chmod 400 MyKey.pem`	Restrict the key file (permissions: Day 11).
`ssh -i MyKey.pem ubuntu@PUBLIC_IP`	Connect as user ubuntu. Replace PUBLIC_IP with your instance address, e.g. 13.233.10.25.

Output (what students will see)

```
The authenticity of host '13.233.10.25' can't be established.
Are you sure you want to continue connecting (yes/no)? yes   <-- first time
Welcome to Ubuntu 22.04 LTS (GNU/Linux 6.8.0-aws x86_64)
ubuntu@ip-172-31-5-20:~$        <-- you are inside the server
```

Windows: Windows 10/11: open PowerShell and use the same command with the full path: `ssh -i C:\Users\YourName\Downloads\MyKey.pem ubuntu@PUBLIC_IP`. For **PuTTY**, first convert the .pem file to .ppk with PuTTYgen.

Operating system	Login user
Ubuntu	ubuntu
Amazon Linux, Red Hat	ec2-user
Debian	admin

If the connection fails

Message	Likely cause and fix
Permissions 0644 ... are too open	Run chmod 400 MyKey.pem.
Connection timed out	Security group has no SSH rule for your current IP (your IP may have changed), the instance has no public IP, or it is not running.
Permission denied (publickey)	Wrong user name (use ubuntu for Ubuntu) or the wrong key file.
Host identification has changed	The IP was reused by a new instance. Run ssh-keygen -R PUBLIC_IP and try again.
Connection refused	The SSH service on the server is not running.

6. First Commands on Your New Server

COPY AND PASTE: Look around the server

```
whoami
hostname
cat /etc/os-release | head -2
nproc
free -h
df -h /
```

Command	What it does (in simple words)
`whoami ; hostname`	Your user (ubuntu) and the server name (contains the private IP).
`cat /etc/os-release | head -2`	Ubuntu version.
`nproc`	Number of CPUs: 1 on a t2.micro.
`free -h`	Memory: about 1 GiB.
`df -h /`	Disk size: about 8 GiB (the EBS volume).

Output (example)

```
ubuntu
ip-172-31-5-20
PRETTY_NAME="Ubuntu 22.04 LTS"
1                                   <-- 1 vCPU
Mem:   957Mi  ...                   <-- about 1 GiB RAM
/dev/root  7.6G  1.9G  5.7G  25% /   <-- 8 GiB disk
```

Ask AWS about this instance (instance metadata)

Every instance can ask a special local address about itself. This works only **on the EC2 server**.

COPY AND PASTE: Instance ID and public IP from inside the server

```
URL=http://169.254.169.254/latest
H=X-aws-ec2-metadata-token
TOKEN=$(curl -s -X PUT -H "$H-ttl-seconds: 300" $URL/api/token)
curl -s -H "$H: $TOKEN" $URL/meta-data/instance-id; echo
curl -s -H "$H: $TOKEN" $URL/meta-data/public-ipv4; echo
```

Command	What it does (in simple words)
`TOKEN=$(curl ... /api/token)`	Gets a short-lived token (IMDSv2, the secure method AWS uses by default).
`curl ... /meta-data/instance-id`	Prints the instance ID, like i-0abc123def456789a.
`curl ... /meta-data/public-ipv4`	Prints the public IP address.

Demo: run a web server (and see why security groups matter)

COPY AND PASTE: Install nginx

```
sudo apt update
sudo apt install -y nginx
systemctl is-active nginx
curl -sI localhost | head -1
```

Output (example)

```
active                  <-- nginx is running
HTTP/1.1 200 OK         <-- it answers on the server itself
```

Now open **http://PUBLIC_IP** in your browser. It will **time out**, because the security group does not allow port 80 yet. In the console, open the instance, then **Security**, the security group, **Edit inbound rules**, and add **HTTP (port 80)** from **Anywhere**. Reload the page: you see **Welcome to nginx!**. (Use this only for practice, then remove the rule.)

Lesson: The server was healthy all the time. The **security group** was the firewall that blocked the web request. Always check the security group when a connection fails.

7. Manage the Instance and Save Money

Action	What happens	Public IP
Reboot	Restarts the OS. The instance stays.	Same
Stop	Shuts down. The disk (EBS) is kept. No compute charge, but you pay for the disk.	Changes when started again (unless Elastic IP)
Start	Starts a stopped instance.	New public IP
Terminate	Deletes the instance. The root disk is deleted by default. **Cannot be undone.**	Released

Avoid surprise bills: Cost safety checklist: (1) Stop or terminate instances after practice. (2) Terminate, not just stop, if you no longer need it, so you stop paying for the disk. (3) Release unused **Elastic IPs**. AWS now charges for public IPv4 addresses, so check pricing. (4) Set a **budget alert**. (5) Check the **Billing** page and the region you used.

Best practices

- Restrict SSH to **your IP only**. Never leave port 22 open to everyone.
- Keep the **.pem key private**. Never share it or upload it to GitHub.
- Use an **Elastic IP** only when you need a fixed public IP.
- Use **tags** (like Name) to find and track your resources.

8. Quick Quiz and Key Takeaways

Question	Answer
What is the AWS Management Console?	The web interface to manage AWS services, usage, billing and security.
Region vs Availability Zone?	A region is a geographical area. An AZ is a separate data center inside a region.
Why deploy in more than one AZ?	If one AZ fails, the app stays available.
What does EC2 provide?	Scalable virtual servers (instances) in the cloud.
Name three EC2 pricing models.	On-Demand, Reserved Instances / Savings Plans, Spot.
What is an AMI? A key pair?	AMI = OS template. Key pair = private .pem and public key used for login.
Which port does SSH use?	22.
Write the SSH command.	ssh -i MyKey.pem ubuntu@PUBLIC_IP (after chmod 400 MyKey.pem).

Question	Answer
Stop vs Terminate?	Stop keeps the disk and can restart. Terminate deletes the instance.

- The AWS Console is the central place to manage your cloud resources.
- Regions and Availability Zones give resilience, low latency and flexibility.
- EC2 launches virtual servers with the size and OS you choose.
- Security groups and key pairs protect access: SSH from your IP only.